Lab

Time: 60 minutes

Write a Matlab script code to:

1. Estimate the noise in the 'NoisyImage.jpg' and denoise it by a suitable filter. Show noisy and restoration image. (**3 point**)
2. Write a Matlab script to perform: **(3 points)**
   a. Read a gray image to matrix X. Create a frame Y with the same size of X, border =50. Combine X and Y such that we have a photo frame with the border is:
      - Black
      - White
   b. Repeat the same task with a color image and color border (black, white, red, green, blue and yellow (R=255, G=255, B=0)).
3. Write a Matlab script code to: **(4 points)**
   a. Read image 'coins.png'. Show the original image and its histogram. (1)
   b. Chose the best threshold based on the histogram and segment. Display the result and the threshold corresponding (1)
   c. Count the number of each type of coins in the image. (1)
   d. Chose one of the two options:

Option 1: Do these questions above with the image “coin2.png” (1 points)

Option 2: Count the number of circles in the image “F13.bmp”. (1 points)